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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B. Tech (CBCS) Degree Examinations

May 2025

BT2204-1 – NANOTECHNOLOGY

(For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What did Gerd Binnig and Heinrich Rohrer create in 1981?**
A) Bucky Balls
B) Scanning Tunneling Microscopy
C) Scanning Electron Microscopy
D) Scanning Probe Microscopy
2. **Which technique involves deposition by chemical reactions?**
A) CVD
B) PVD
C) Sol Gel Method
D) VLD
3. **Which among the following is an example of using nanotechnology in healthcare?**
A) Quantum dots for imaging
B) Quantum dots in quantum cryptography
C) Quantum dots as sensors
D) Quantum dots in LED
4. **Which ancient material exhibits nanotechnology properties due to dichroic glass?**
A) Greek Amphora
B) Egyptian Sarcophagus
C) Mesopotamian Figurines
D) Roman Cage Cup
5. **What does "DPN" stand for in nanofabrication?**
A) Direct Process Nanotechnology
B) Dense Particle Nanofabrication
C) Diffraction Pattern Nanolithography
D) Dip Pen Nanolithography
6. **Which lithography uses self-assembled monolayers of spheres as masks?**
A) Photolithography
B) UV Lithography
C) Nanosphere Lithography
D) E-Beam Lithography
7. **Nanowires are primarily used in applications involving:**
A) Chemical catalysis
B) Magnetic shielding
C) Optical coatings
D) Electrical conduction
8. **Which technique provides information about chemical bonds in materials?**
A) Raman Spectroscopy
B) X-ray Diffraction
C) Optical Spectroscopy
D) Scanning Electron Microscopy
9. **What technique is commonly used for cell trapping in microfluidic devices?**
A) Piezoelectric sorting
B) Magnetic levitation
C) Optical trapping
D) Dielectrophoresis
10. **What is the key application of magnetic microbeads in microfluidic devices?**
A) Biochemical detection
B) Filtering air
C) Heating samples
D) Mixing fluids
11. **The Coanda effect in micromixers is used to:**
A) Reduce turbulence
B) Enhance diffusion
C) Direct fluid flow along a curved surface
D) Increase fluid resistance

12. What is the primary mechanism used by passive micromixers to mix fluids?
 A) Diffusion and chaotic advection B) Mechanical stirring
 C) Electro-osmosis D) Magnetic agitation
13. Which of the following is NOT a common application of micropumps?
 A) Drug delivery B) Chemical analysis
 C) Lab-on-a-chip devices D) Satellite fuel propulsion
14. In soft lithography, which material is used as an elastomer for microfluidic devices?
 A) PDMA B) Glass
 C) Gold D) Silicon
15. In MEMS fabrication, photolithography involves:
 A) Depositing thin films B) Selectively exposing photoresist to light
 C) Bonding wafers together D) Growing silicon crystals
16. A sensor is a device that:
 A) Stores energy B) Detects environmental changes
 C) Acts as a power source D) Converts one form of energy to another
17. Which type of sensor measures temperature?
 A) Thermocouple B) Piezoelectric Sensor
 C) Accelerometer D) Pressure Sensor
18. A biosensor detects:
 A) Only physical changes B) Magnetic fields
 C) Only mechanical stress D) Biological components like proteins and cells
19. Which of the following is an example of a mechanical sensor?
 A) Pressure sensor B) pH electrode
 C) Optical transducer D) Infrared detector
20. Which bioelectric signal is NOT commonly measured in modern patient care?
 A) MRI B) ECG
 C) EMG D) EEG

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is the scope of Nanotechnology? Explain	8	L*2	1	1
	b) Define dendrimers. Explain their structure, properties and applications.	8	L2	2	1
2.	a) Explain the structure property relationship in nanomaterials?	8	L1	1	1
	b) What is the principle of Atomic Force Microscopy. Explain its working with a neat diagram	8	L2	2	5
3.	a) Define photolithography. What is a photoresist? Explain its types and working with a neat diagram.	8	L2	1	5
	b) How does XRD analyse the crystalline structure of nanomaterials? Explain.	8	L2	2	1
Unit – II					
4.	a) Explain the use of nanotechnology in delivery of small molecules like proteins in the body.	8	L1	3	2
	b) What is microfluidics? Explain its components.	8	L2	4	1
5.	a) Liposomes are used in anticancer drug delivery. Explain.	8	L1	3	2
	b) Write a note on Magnetic Micro/Nano-Bead-Based Biochemical Detection System.	8	L1	4	5
6.	a) How is Nanotechnology used in drug development? Discuss.	8	L2	3	2
	b) How are microchips used for genomic and proteomic analysis? Explain.	8	L2	4	5

Unit – III

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| 7. | a) | What is a mechanical transducer? How is it used to measure temperature? | 8 | L2 | 5 | 1 |
| | b) | Give an account of surgical applications of MEMS. | 8 | L1 | 5 | 1 |
| 8. | a) | Explain the use of MEMS in drug delivery. | 8 | L2 | 5 | 1 |
| | b) | Give an account of optical and micro-cantilever-based sensors in diagnostics. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
